

S&P 500 hedging: was protection worth buying?

S&P 500 via SPY options · decision date 1 April 2025

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30D ATM VOL, 1 APR	RECOMMENDED HEDGE	COST, % OF NAV	COLLAR SOLD AT VIX $\geq$ 40
17.7%	Collar 539/583	0.25%	+\$33.2M

01 Summary

On 2 April 2025, after the close, the United States announced reciprocal tariffs on most of its trading partners. In the next four sessions the S&P 500 fell 12.1% and the VIX closed at 52.3 on 8 April. On 9 April a 90-day pause lifted the index 9.5% in a day, and by 30 May SPY was back at 589.39, above its 1 April level.

This memo asks what a portfolio manager running a **\$500M US equity book** (beta 1) should have done on the evening of 1 April. It rebuilds the S&P 500 implied volatility surface for every trading day of the first half of 2025 from real SPY option quotes, checks it against a VIX it never sees, prices four hedges at the 1 April quotes and follows them to expiry. Protection was not expensive: 30-day ATM vol was 17.7%, below the 20.1% realised over the previous month. The **539/583 collar** cut the drawdown from -11.8% to **-3.4%** for 0.25% of NAV. Held to expiry every hedge lost money; sold at the first close with the VIX at or above 40, the collar finished **\$33.2M ahead** of the unhedged book.

CONCLUSION

Buy the 539/583 collar and write the exit into the trade on day one: sell it at the first close with the VIX at or above 40. In April 2025 the rule for taking profit decided the result more than the choice of strikes.

02 The volatility surface

Each day's chain has three expiries (about 2, 4 and 8 weeks). The underlying is inferred from put-call parity, because the quotes are not always taken at the close. Implied vols come from Black-76 on the forward. Each expiry gets a raw SVI smile fitted to the quotes, with penalties that rule out butterfly and calendar arbitrage. Expiries are joined in total variance; beyond the listed range the level is anchored to the Cboe VIX9D, VIX3M and VIX6M. The 30-day VIX is left out, as a test.

CHECK	RESULT
Data	122 trading days, 2 Jan to 30 Jun 2025; 6,166 out-of-the-money SPY quotes fitted (bid and ask)
Fit error	0.47 vol points RMSE on average, median 0.41, worst day 1.16
VIX rebuilt out of sample	mean error -0.15 vol points, standard deviation 0.63, correlation 0.996
Butterfly arbitrage	0 violations on 3.4 million grid points
Calendar arbitrage	0.015% of grid points, all in the far call wing beyond +17%, largest breach 4×10^{-5} in total variance

The worst VIX gap is on 8 April (49.4 rebuilt vs 52.3): SPY options stand in for SPX, and the deepest puts behind the index are not in the dataset. Implied vols match the data vendor's to a median of 0.02 vol points.

03 Was protection expensive on 1 April?

The case for hedging rests on price as much as on the event. On 1 April the surface said protection was fairly priced. At-the-money vol was above its half-year median but below what the market had just realised, and the skew was ordinary. The one clear signal was in the short end: the 9-day variance index stood above the 30-day and 3-month ones, so the market was already paying for a jump in the next days, not for a slow decline.

METRIC	1 APRIL	H1 2025 MEDIAN	READING
30-day ATM implied vol	17.7%	15.5%	above the median, below realised
21-day realised vol, S&P 500	20.1%	16.2%	elevated by the March correction
Implied minus realised	-2.4	-1.4	options priced below recent moves
25-delta risk reversal, 30d	-4.96	-4.50	put skew in line with the half-year
90% put minus ATM, 30d	+9.51	+10.05	slightly flatter than usual
VIX9D / VIX / VIX3M	24.5 / 21.8 / 21.8	18.4 / 19.1 / 20.7	front end inverted: the event was priced
Cboe SKEW	143	142	tail demand in line

04 Four structures on the 30 May expiry

All four use listed SPY strikes on the 30 May 2025 expiry (59 days) and are priced at the real 1 April quotes: the ask for options bought, the bid for options sold. 8,907 contracts cover the book at SPY 561.36. The model mid comes from the fitted surface.

STRUCTURE	LEGS	PREMIUM / SH	MODEL MID	COST	% NAV	PROTECTION
Long put 539	+539P	\$8.36	\$8.08	\$7.45M	1.49%	Below 539 (-4.0%)
Put spread 539/475	+539P -475P	\$6.59	\$6.21	\$5.87M	1.17%	539 to 475 (-4% to -15%)
Collar 539/583	+539P -583C	\$1.40	\$1.26	\$1.25M	0.25%	Below 539; upside capped at 583 (+3.9%)
Put-spread collar 539/475/583	+539P -475P -583C	-\$0.37	-\$0.61	-\$0.33M	-0.07%	539 to 475; capped at 583

The collar is the most efficient: the 583 call pays for most of the 539 put, so the book keeps nearly all of the outright put's protection for a sixth of its cost (0.25% vs 1.49% of NAV). The price is the upside above 583. The put spreads cost less than the outright put but stop paying below 475, a level SPY never reached: its lowest close was 496.48 on 8 April and its intraday low 481.80 on 7 April.

05 What happened

STRUCTURE	NAV, 8 APR	MAX DD, HELD	NAV 30 MAY, HELD	SOLD 4 APR FOR	NAV 30 MAY, SOLD	SOLD VS UNHEDGED
Unhedged	\$443.8M	-11.8%	\$524.4M	–	\$524.4M	–
Long put 539	\$478.6M	-4.6%	\$516.9M	\$36.1M	\$553.0M	+\$28.6M
Put spread 539/475	\$463.4M	-7.7%	\$518.5M	\$22.6M	\$541.1M	+\$16.8M
Collar 539/583	\$483.8M	-3.4%	\$518.0M	\$34.5M	\$557.6M	+\$33.2M
Put-spread collar 539/475/583	\$468.6M	-6.5%	\$519.6M	\$21.1M	\$545.7M	+\$21.4M

Hedges marked every day on that day's fitted surface. "Sold" closes every leg at the first close with the VIX at or above 40 (4 April, VIX 45.3), at model mid less half the quoted bid-ask spread.

At the 8 April low the unhedged book was worth \$443.8M. The collar held it at \$483.8M and the outright put at \$478.6M. Then the market turned: after the pause and the US-China truce of 12 May, SPY finished at 589.39, so the puts expired worthless and the collars had to pay out on the short 583 call. **Held to expiry, every structure ended below the unhedged book**, the collar by \$6.3M.

Selling on the VIX trigger changed the outcome. On 4 April the collar was worth \$34.5M; banking it and staying long the index left the book **\$33.2M ahead** on 30 May. The rule sold two sessions before the low and gave up \$6.8M of the collar's peak value, but it did not depend on calling the bottom.

06 Recommendation

- **Buy the 539/583 collar on 1 April** for 0.25% of NAV: the best protection per dollar among the four, with a drawdown of -3.4%.
- **Pre-commit the exit.** Close the collar at the first close with the VIX at or above 40 rather than holding it to expiry.
- **Keep marking the hedge on the surface.** A daily mark at a known model price, checked against quotes, is what makes a rule like $VIX \geq 40$ executable when markets are moving 5% a day.

07 Limitations

- SPY options are American and SPY is not the index; the Cboe indices are computed on SPX. Early exercise matters little for out-of-the-money options, but the VIX comparison carries a basis.
- Three expiries a day: tenors shorter than about 2 weeks or longer than about 8 weeks come from the $\sqrt{T}$ rule and the Cboe anchors, not from quotes.
- Hedges are marked on the model; on the 12 occasions the exact strikes were quoted, model and market mids differ by \$0.13 on average.
- The book is assumed to track SPY one-for-one. The VIX-40 rule is one choice among many and the result depends on the path the market took in April 2025.

Sources. SPY option chains and prices: DoltHub (post-no-preference/options, /stocks). VIX, VIX9D, VIX3M, VIX6M, SKEW and S&P 500: Cboe. Treasury yields: FRED. Method: Gatheral & Jacquier (2014), Lee (2004), Breeden & Litzenberger (1978), Cboe VIX white paper. Event dates from contemporary press (CNBC, NPR). Educational project, not investment advice.